PAC-Bayesian Model Averaging

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Abstract

PAC-Bayesian learning methods combine the informative priors of Bayesian methods with distribution-free PAC guarantees. Building on earlier methods for PAC-Bayesian model selection, this paper presents a method for PAC-Bayesian model averaging. The method constructs an optimized weighted mixture of concepts analogous to a Bayesian posterior distribution. Although the main result is stated for bounded loss, a preliminary analysis for unbounded loss is also given.

1 INTRODUCTION

A PAC-Bayesian approach to machine learning attempts to combine the advantages of both PAC and Bayesian approaches [12, 8]. The Bayesian approach has the advantage of using arbitrary domain knowledge in the form of a Bayesian prior. The PAC approach has the advantage that one can prove guarantees for generalization error without assuming the truth of the prior. A PAC-Bayesian approach combines the features of the PAC and Bayesian approaches — it bases the bias of the learning algorithm on an arbitrary prior distribution, thus allowing the incorporation of domain knowledge, and yet provides a guarantee on generalization error that is independent of any truth of the prior.

PAC-Bayesian approaches are related to structural risk minimization (SRM) [6]. Here we interpret this broadly as describing any learning algorithm optimizing a tradeoff between the "complexity", "structure", or "prior probability" of the concept or model and the "goodness of fit", "description length", or "likelihood" of the training data. Under this interpretation of SRM, Bayesian algorithms which select a concept of maximum posterior probability (MAP algorithms) are viewed as a kind of SRM algorithm. Various approaches to SRM are compared both theoretically and experimentally by Kearns et al. in [6]. They give experimental evidence that Bayesian and MDL algorithms tend to over-fit in experimental settings where the Bayesian assumptions fail. A PAC-Bayesian approach uses a prior distribution analogous to that used in MAP or MDL but provides a theoretical guarantee against over-fitting independent of the truth of the prior.

Earlier work on PAC-Bayesian algorithms has focused on model selection — selecting either a single concept or a uniformly weighted set of concepts. Here we consider nonuniform model averaging, i.e., selecting a weighted mixture of the concepts.

Model averaging is empirically important in certain applications. For example, in statistical language modeling for speech recognition one "smooths" a trigram model with a bigram model and smooths the bigram model with a unigram model. This smoothing is essential for minimizing the cross entropy between, say, the model and a test corpus of newspaper sentences. It turns out that smoothing in statistical language modeling is more naturally formulated as model averaging than as model selection. A smoothed language model is very large — it contains a full trigram model, a full bigram model and a full unigram model as parts. If one uses MDL to select the structure of a language model, selecting model parameters with maximum likelihood, the resulting structure is much smaller than that of a smoothed trigram model. Furthermore, the MDL model performs quite badly. However, a smoothed trigram model can be theoretically derived as a compact representation of a Bayesian mixture of an exponential number of (smaller) suffix tree models [10].

Model averaging can also be applied to decision trees. A common method of constructing decision trees is to first build an overly large tree which over-fits the training data and then prune the tree in some way so as to get a smaller tree that does not over-fit the data [11, 5]. An alternative to pruning is to construct a weighted mixture of the subtrees of the original over-fit tree. It is possible to construct a concise representation of a weighting over exponentially many different subtrees [3, 9, 4].

This paper proves a new PAC-Bayesian theorem giving a bound on the generalization error of weighted mixtures. A weighted mixture which gives too much weight to models with low prior probability will over-fit the

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data in the sense that its provable loss bound is large. A weighting that puts too much weight on models that fit the training data poorly but have high prior probability under-fits the data in the sense that its provable loss bound is also large. A weighting that minimizes the provable loss bound is formally analogous to the classical Bayesian posterior distribution.

The main result of this paper are for bounded loss function such as classification error rates. However, section 5 briefly considers the case of unbounded loss such as log loss. By allowing for classical log loss, the model selection criterion and PAC-Bayesian posterior distribution suggested in section 5 provide a tighter connection between the PAC-Bayesian and classical Bayesian approaches.

2 Statement of Main Results

In this section we state the main result of this paper. Proofs are given in later sections. The main result of this paper (theorem 1) holds for arbitrary (possibly uncountable) concept classes. However, to simplify the exposition, here we consider only countable concept classes. We assume a prior probability distribution P on a countable concept class C with concepts c_1, c_2, c_3, ... where c_i has prior probability P_i, and a sampling distribution D on a countable set of instances x_1, x_2, x_3, ... such that the probability that sampling selects x_i is D_i. We also assume a loss function l such that for any concept c and instance x we have $l(c, x) \in [0, 1]$ (unbounded loss functions are considered in section 5). For example, we might have that concepts are predicates on instances and there is a target concept c_t such that $l(c, x)$ is 0 if $c(x) = c_t(x)$ and 1 otherwise. We define $l(c)$ to be the expectation over sampling an instance x of $l(c,x)$, i.e., $\sum D_i \, l(c, x_i)$. We let S range over samples of m instances each drawn independently according to distribution D. We define $\hat{l}(c, S)$ to be $\frac{1}{m}\sum_{x\in S} l(c, x)$. If Q is a probability distribution on concepts where c_i has probability Q_i then $l(Q)$ denotes $\sum Q_i l(c_i)$ and $\hat{l}(Q, S)$ denotes $\sum Q_i \hat{l}(c_i, S)$. The notation $\forall^\delta S \; \Phi(S)$ signifies that the probability over the generation of the sample S of $\Phi(S)$ is at least $1-\delta$.

Before describing PAC-Bayesian model averaging we review PAC-Bayesian model selection. The departure point for the current paper is the following model selection "folk theorem" from [8].[1]

$$\forall^\delta S \;\; \forall c_i \;\; l(c_i) \leq \hat{l}(c_i, S) + \sqrt{\frac{\ln\frac{1}{P_i} + \ln\frac{1}{\delta}}{2m}} \tag{1}$$

This theorem states that for most draws of a sample S we have that, for all concepts c_i, the generalization loss of c_i is bounded by the measured loss of c_i plus a penalty term that gets larger as the prior probability of c_i declines. The bound yields a model selection algorithm where one selects a concept c^* so as to minimize the upper bound.

[1]Results similar to (1) have appeared repeatedly in the literature, e.g., [1, 2, 7].

The proof of (1) is straightforward. By the Chernoff bound we have that for any fixed concept c_i the probability that c_i violates the theorem is no larger than $P_i\delta$. By the union bound the probability that some c_i violates the theorem is then no larger than $\sum P_i\delta = \delta\sum P_i = \delta$.

Formula (1) can be related to Bayesian model selection or MDL. We assume that each concept has an associated distribution on instances and let $P(x|c)$ denote the probability of x under the distribution associated with c. We let $L(c, x)$ denote the classical log loss of c on x, i.e., the quantity $\ln\frac{1}{P(x|c)}$. For a sample S of size m we define $\hat{L}(c, S)$ to be $\frac{1}{m}\sum_{x\in S} L(c, x)$. The concept maximizing the Bayesian posterior is the same as the concept minimizing $\ln\frac{1}{P_i P(S|c_i)}$ which is the same as the concept minimizing $\hat{L}(c_i, S) + \frac{\ln\frac{1}{P_i}}{m}$. In the MDL interpretation $\frac{\ln\frac{1}{P_i}}{m}$ is proportional to the complexity of the concept c_i and $\hat{L}(c_i, S)$ is proportional to the complexity of the sample given c_i. Formula (1) defines a different trade-off between the "concept complexity" $\frac{\ln\frac{1}{P_i}}{m}$ and the "sample complexity" $\hat{l}(c_i, S)$.

The main result in [8] is a generalization of (1) to a uniform statement over all *subsets* of the concepts. The main result of this paper is a generalization of (1) to a uniform statement over *distributions* on concepts. The bound involves the Kullback-Leibler divergence, denoted $D(Q||P)$, from distribution Q to distribution P. The quantity $D(Q||P)$ is defined to be $\sum Q_i \ln\frac{Q_i}{P_i}$.

To simplify the mathematical analysis the theorem is restricted to distributions Q which have the property that for each i, either $Q_i \geq P_i$ or $Q_i = 0$. Such distributions will be called *pruned* — if $Q_i < P_i$ then the concept c_i is pruned away by setting $Q_i = 0$ and renormalizing the remainder. The restriction to pruned distributions is a technicality which should have no significance in practice. To see this consider statistical language modeling or decision tree learning. In each case the Bayesian posterior is typically dominated by models whose size grows in proportion to the size of the training data. In a language model, for example, the number of trigrams in typical suffix tree model is proportional to the size of the training data. Such large models have extremely small prior probability. In order for these models to dominate the posterior we must have $Q_i \gg P_i$ for these dominant models. Hence pruning the posterior will have very little affect on the posterior probability of the dominant models.

Theorem 1 *With probability at least* $1-\delta$ *over the selection of the sample* S *of size* m *we have the following for all pruned distributions* Q.

$$l(Q) \leq \hat{l}(Q, S) + \sqrt{\frac{D(Q||P) + \ln\frac{1}{\delta} + \frac{5}{2}\ln m + 8}{2m-1}}$$

In the case where Q_i is concentrated on the single concept c_i the quantity $D(Q||P)$ equals $\ln\frac{1}{P_i}$ and, for large m, theorem 1 is essentially the same as (1). The main result in [8] can also be viewed as a special case of theorem 1. Theorem 1 corresponds to a model averaging learning algorithm where one selects the distribution Q^* minimizing the bound. The optimal distribution Q^* can be viewed as a PAC-Bayesian posterior distribution analogous to the classical Bayesian posterior.

Theorem 1 can be related to classical Bayesian model averaging by again considering classical log loss. It is possible to show that the classical Bayesian posterior distribution is the unique distribution Q minimizing $\hat{L}(Q, S) + \frac{D(Q||P)}{m}$.[2] Theorem 1 defines a different trade-off between $\hat{l}(Q, S)$ and $\frac{D(Q||P)}{m}$.

Theorem 1 is proved from the following lemma.

Lemma 2 *For $\ln\frac{1}{\delta} \leq 2m$ we have that with probability $1-\delta$ over the selection of a sample S of size m the following holds for all pruned distributions satisfying $D(Q||P) \leq 2m$.*

$$l(Q) \leq \hat{l}(Q, S) + \sum Q_i \sqrt{\frac{\ln\frac{Q_i}{P_i} + \ln\frac{1}{\delta} + \frac{5}{2}\ln m + 8}{2m-1}}$$

Lemma 2 is proved in section 3. Note that if $\ln\frac{1}{\delta} > 2m$ or $D(Q||P) > 2m$ then the bound in theorem 1 follows from the fact that $l(Q) \leq 1$. If $\ln\frac{1}{\delta} \leq 2m$ and $D(Q||P) \leq 2m$ then theorem 1 follows from lemma 2 by an application of Jensen's inequality. Assuming that we are only interested in cases where the premises of lemma 2 hold, lemma 2 gives a tighter bound than theorem 1.

Section 4 argues that the distribution optimizing the bound in lemma 2 is approximately equal to the following where $\hat{l}(c_i, S)$ is abbreviated as $\hat{l}_i$.

$$Q_i^* \approx \begin{cases} 0 & \text{if } \hat{l}_i > z \\ P_i\alpha & \text{if } \hat{l}_i = z \\ P_i e^{2m(z-\hat{l}_i)^2} & \text{if } \hat{l}_i < z \end{cases} \tag{2}$$

In (2) the parameters z and α are the unique values in $[0, 1]$ such that $\sum Q_i = 1$. An explicit calculation of these parameters is given in section 4.

It is interesting to compare (2) to the Bayesian posterior. The classical posterior probability of c_i can be written as $P_i e^{m(z-\hat{L}_i)}$ where z is selected to normalize the distribution. This is somewhat similar the last line in (2).

[2]To show this it suffices to show that $\hat{L}(Q, S) + \frac{D(Q||P)}{m}$ is a function of $D(Q||R)$ where R is the classical posterior:

$$\begin{aligned} \hat{L}(Q, S) + \frac{D(Q||P)}{m} &= \sum Q_i \frac{1}{m} \sum_{x\in S} \ln\frac{1}{P(x|c_i)} + \frac{D(Q||P)}{m} \\ &= \frac{1}{m}\left[\sum Q_i \ln\frac{1}{P(S|c_i)} + \sum Q_i \ln\frac{Q_i}{P_i}\right] \\ &= \frac{1}{m}\sum Q_i \ln\frac{Q_i}{P_i P(S|c_i)} \\ &= \frac{1}{m}\sum Q_i(\ln\frac{Q_i}{R_i} - \ln Z) \\ &= \frac{1}{m}(D(Q||R) - \ln Z) \end{aligned}$$

3 Proof of Lemma 2

The departure point for the proof of lemma 2 is the following where S is a sample of size m and γ_i abbreviates $|l(c_i) - \hat{l}(c_i, S)|$.

Lemma 3

$$\forall^\delta S \quad \sum P_i e^{(2m-1)\gamma_i^2} \leq \frac{4m}{\delta}$$

Proof: It suffices to prove the following.

$$E_S\ E_{c_i}\ e^{(2m-1)\gamma_i^2} \leq 4m \tag{3}$$

Lemma 3 follows from (3) by an application of Markov's inequality. To prove (3) it suffices to prove the following for any individual given concept.

$$E_S\ e^{(2m-1)\gamma^2} \leq 4m \tag{4}$$

By the Chernoff bound we have $P(\gamma \geq x) \leq 2e^{-2mx^2}$. We now consider the density function $f(\gamma)$ maximizing $\int_0^\infty e^{(2m-1)\gamma^2} f(\gamma)d\gamma$ subject to the constraint that $\int_x^\infty f(\gamma)d\gamma \leq 2e^{-2mx^2}$. The maximum occurs when we have $\int_x^\infty f(\gamma)d\gamma = 2e^{-2mx^2}$ which is realized when $f(\gamma) = 8m\gamma e^{-2m\gamma^2}$. So we have the following

$$\begin{aligned} E_S\ e^{(2m-1)\gamma^2} &\leq \int_0^\infty e^{(2m-1)\gamma^2} f(\gamma)d\gamma \\ &= \int_0^\infty 8m\gamma e^{(2m-1)\gamma^2} e^{-2m\gamma^2} d\gamma \\ &= \int_0^\infty 8m\gamma e^{-\gamma^2} d\gamma \\ &= 4m \end{aligned}$$

□

To prove lemma 2 we consider selecting a sample S. Lemma 3 implies that with probability at least $1-\delta$ we have the following.

$$\sum P_i e^{(2m-1)\gamma_i^2} \leq \frac{4m}{\delta} \tag{5}$$

After selecting S satisfying (5) we consider an arbitrary "posterior" distribution Q. Lemma 2 can be viewed as an upper bound on the quantity $l(Q) - \hat{l}(Q, S)$. But note that $l(Q) - \hat{l}(Q, S) = \sum Q_i(l(c_i) - \hat{l}(c_i, S)) \leq \sum Q_i\gamma_i$. So to prove lemma 2 it now suffices to show that (5) plus $\ln\frac{1}{\delta} \leq 2m$ implies the following for all pruned distributions Q such that $D(Q||P) \leq 2m$.

$$\sum Q_i\gamma_i \leq \sum Q_i \sqrt{\frac{\ln\frac{Q_i}{P_i} + \ln\frac{1}{\delta} + \frac{5}{2}\ln m + 8}{2m-1}} \tag{6}$$

To prove (6) for a given Q we select γ_i so as to maximize the quantity $\sum Q_i\gamma_i$ subject to the constraint (5). Using Lagrange multipliers we set the gradient of the constraint to be equal to a multiplier λ times the gradient of the objective function.

$$P_i 2(2m-1)\gamma_i e^{(2m-1)\gamma_i^2} = \lambda Q_i$$

Formula (6) is trivially true if $Q_i > 0$ but $P_i = 0$ for some i. So we can assume without loss of generality that $P_i > 0$ whenever $Q_i > 0$. This allows the above to be rewritten as follows.

$$2(2m-1)\gamma_i e^{(2m-1)\gamma_i^2} = \frac{\lambda Q_i}{P_i}$$

Note that $2(2m-1)xe^{(2m-1)x^2}$ is an unbounded monotonically increasing function of x. We now define $\Delta_i(\lambda)$ to be the unique non-negative value satisfying the following.

$$2(2m-1)\Delta_i(\lambda)e^{(2m-1)\Delta_i^2(\lambda)} = \frac{\lambda Q_i}{P_i} \tag{7}$$

Now note that $\sum P_i e^{(2m-1)\Delta_i^2(\lambda)}$ is an unbounded monotonically increasing function of λ. We now define λ^* to be the unique nonnegative value such that we have the following.

$$\sum P_i e^{(2m-1)\Delta_i^2(\lambda^*)} = \frac{4m}{\delta} \tag{8}$$

Note that $\Delta_i(0) = 0$ and $\sum P_i e^{(2m-1)\Delta_i^2(0)} = 1 < \frac{4m}{\delta}$. So we must have $\lambda^* > 0$ and hence $\Delta_i(\lambda^*) > 0$ for $Q_i > 0$.

Lemma 4 *For any values of γ_i satisfying (5) we have $\sum Q_i\gamma_i \leq \sum Q_i\Delta_i(\lambda^*)$.*

Proof: If $\gamma_i < 0$ then replacing γ_i by $-\gamma_i$ increases $\sum Q_i\gamma_i$ without violating (5). Hence we can assume without loss of generality that $\gamma_i \geq 0$. Also, note that if $\sum P_i e^{(2m-1)\gamma_i^2} < \frac{4m}{\delta}$ then $\sum Q_i\gamma_i$ can be increased by increasing some γ_i. Hence we can assume without loss of generality that $\sum P_i e^{(2m-1)\gamma_i^2} = \frac{4m}{\delta}$. Also, if $\gamma_j > 0$ for some j such that $Q_j = 0$ then $\sum Q_i\gamma_i$ can be increased by setting $\gamma_j = 0$ while increasing some other γ_k with $Q_k > 0$. Hence we can assume without loss of generality that $\gamma_j = 0$ whenever $Q_j = 0$. Finally, if there exist two indices j and k with $Q_j > 0$ and $Q_k > 0$ and $\frac{P_j}{Q_j}\gamma_i e^{(2m-1)\gamma_j^2} > \frac{P_k}{Q_k}\gamma_j e^{(2m-1)\gamma_k^2}$ then $\sum Q_i\gamma_i$ can be increased by increasing γ_k and decreasing γ_j while holding $\sum P_i e^{(2m-1)\gamma_i^2}$ constant. So we can assume without loss of generality that there exists a value λ' such that for all indices i with $Q_i > 0$ we have $2(2m-1)\gamma_i e^{(2m-1)\gamma_i^2} = \frac{\lambda' Q_i}{P_i}$. This implies that $\gamma_i = \Delta_i(\lambda')$. But we then have that $\sum P_i e^{(2m-1)\Delta_i^2(\lambda')} = \frac{4m}{\delta}$ which implies that $\lambda' = \lambda^*$. So we now have that $\gamma_i = \Delta_i(\lambda^*)$ which implies the result. □

It now suffices to bound $\sum Q_i\Delta_i(\lambda^*)$. Equation (7) implies that for $\lambda \gg 1$ and $Q_i \geq P_i$ we have the following

$$\Delta_i(\lambda) \approx \sqrt{\frac{\ln\frac{\lambda Q_i}{P_i}}{2m-1}}$$

This approximate relationship is made more precise in the following two lemmas.

Lemma 5 *For $m \geq 1$, $P_i > 0$, $Q_i \geq P_i$, and $\lambda \geq e$, we have the following*

$$\Delta_i(\lambda) \leq \sqrt{\frac{\ln\frac{\lambda Q_i}{P_i}}{2m-1}}$$

Proof: Let $f(x)$ be the function mapping x to the quantity $2(2m-1)xe^{(2m-1)x^2}$. By definition, $\Delta_i(\lambda)$ satisfies $f(\Delta_i(\lambda)) = \frac{\lambda Q_i}{P_i}$. Note that for $x \geq 0$ we have that $f(x)$ is a monotonically increasing function. Hence for $x \geq 0$ and $f(x) \geq \frac{\lambda Q_i}{P_i}$ we must have $\Delta_i(\lambda) \leq x$. Under the assumptions of the lemma we have $\ln\frac{\lambda Q_i}{P_i} \geq 1$ which implies the following.

$$\begin{aligned} f\left(\sqrt{\frac{\ln\frac{\lambda Q_i}{P_i}}{2m-1}}\right) &= 2\sqrt{(2m-1)\ln\frac{\lambda Q_i}{P_i}}\frac{\lambda Q_i}{P_i} \\ &\geq \frac{\lambda Q_i}{P_i} \end{aligned}$$

□

Lemma 6 *For $m \geq 1$, $P_i > 0$, $Q_i \geq P_i$, and $\lambda \geq e$ we have the following.*

$$\Delta_i(\lambda) \geq \sqrt{\frac{\ln\frac{\lambda Q_i}{P_i} - \frac{1}{2}\ln m - \ln\ln\frac{\lambda Q_i}{P_i} - 2}{2m-1}}$$

Proof: By an argument similar to that in the proof of lemma 5, to show that $\Delta_i(\lambda) \geq x$ it suffices to show that $f(x) \leq \frac{\lambda Q_i}{P_i}$. In particular we have the following.

$$f\left(\sqrt{\frac{\ln\frac{\lambda Q_i}{P_i} - \frac{1}{2}\ln m - \ln\ln\frac{\lambda Q_i}{P_i} - 2}{2m-1}}\right)$$

$$= 2\sqrt{(2m-1)(\ln\frac{\lambda Q_i}{P_i} - \frac{1}{2}\ln m - \ln\ln\frac{\lambda Q_i}{P_i} - 2)}\frac{\lambda Q_i}{P_i}\frac{1}{\sqrt{m}\ln\frac{\lambda Q_i}{P_i}e^2}$$

$$\begin{aligned} &\leq 4\sqrt{m\ln\frac{\lambda Q_i}{P_i}}\frac{\lambda Q_i}{P_i}\frac{1}{\sqrt{m}\ln\frac{\lambda Q_i}{P_i}e^2} \\ &= \frac{\lambda Q_i}{P_i}\frac{1}{\sqrt{\ln\frac{\lambda Q_i}{P_i}}}\frac{4}{e^2} \\ &\leq \frac{\lambda Q_i}{P_i} \end{aligned}$$

□

Lemma 7 *For Q pruned, $D(Q||P) \leq 2m$, and $\ln\frac{1}{\delta} \leq 2m$ we have $\lambda^* \leq \frac{64e^2m^{5/2}}{\delta}$.*

Proof: Let $g(\lambda)$ be the function mapping λ to the quantity $\sum P_i e^{(2m-1)\Delta_i^2(\lambda)}$. The quantity λ^* is defined by $g(\lambda^*) = \frac{4m}{\delta}$. Since g is a monotonically increasing function, $g(x) \geq \frac{2m}{\delta}$ implies $\lambda^* \leq x$. Lemma 6 implies that for $x \geq e$ we have the following.

$$\begin{aligned}
g(x) &= \sum P_i e^{(2m-1)\Delta_i^2(x)} \\
&\geq \sum P_i \frac{xQ_i}{P_i} \frac{1}{\sqrt{m} \ln \frac{xQ_i}{P_i} e^2} \\
&= \frac{x}{e^2\sqrt{m}} \sum \frac{Q_i}{\ln \frac{xQ_i}{P_i}} \\
&\geq \frac{x}{e^2\sqrt{m}} \frac{1}{\sum Q_i \ln \frac{xQ_i}{P_i}} \\
&= \frac{x}{e^2\sqrt{m}(D(Q||P) + \ln x)} \\
&\geq \frac{x}{e^2\sqrt{m}(2m + \ln x)}
\end{aligned}$$

Now inserting $x = \frac{64e^2m^{5/2}}{\delta}$ we get the following.

$$\begin{aligned}
g(\frac{64e^2m^{5/2}}{\delta}) &\geq \frac{64m^2}{\delta(2m + \frac{5}{2}\ln m + \ln\frac{1}{\delta} + 8)} \\
&\geq \frac{64m^2}{\delta(2m + \frac{5}{2}m + 2m + 8m)} \\
&\geq \frac{4m}{\delta}
\end{aligned}$$

□

Lemma 2 now follows from the following calculation.

$$\begin{aligned}
l(Q) - \hat{l}(Q,\, S) &\leq \sum Q_i \gamma_i \\
&\leq \sum Q_i \Delta_i(\lambda^*) \\
&\leq \sum Q_i \Delta_i \left(\frac{64e^2m^{5/2}}{\delta}\right) \\
&\leq \sum Q_i \sqrt{\frac{\ln \frac{64e^2m^{5/2}Q_i}{\delta P_i}}{2m-1}} \\
&\leq \sum Q_i \sqrt{\frac{\ln \frac{Q_i}{P_i} + \frac{5}{2}\ln m + \ln\frac{1}{\delta} + 8}{2m-1}}
\end{aligned}$$

4 A PAC-Bayesian Posterior Distribution

In the case where $\ln\frac{1}{\delta} \leq 2m$ and $D(Q||P) \leq 2m$ lemma 2 provides a tighter bound on $l(Q)$ than does theorem 1. In this section we derive a PAC-Bayesian posterior distribution by approximately solving for the distribution minimizing the bound in lemma 2. We first note that lemma 2 can be written as

$$l(Q) \leq B(Q)$$

where

$$\begin{aligned}
B(Q) &= \sum Q_i(\hat{l}_i + \Gamma_i) \\
\hat{l}_i &= \hat{l}(c_i,\, S) \\
\Gamma_i &= \sqrt{\frac{\ln\frac{Q_i}{P_i} + \alpha}{\beta}} \\
\alpha &= \ln\frac{1}{\delta} + \frac{5}{2}\ln m + 8 \\
\beta &= 2m - 1.
\end{aligned}$$

Section 2 gives an argument that for decision trees and language modeling we have that the posterior distribution is dominated by concepts where $Q_i \gg P_i$. Here we make an even stronger assumption. Note that in most learning applications the training loss is significantly less than the generalization loss. In particular we expect that, for the optimal posterior Q, the bound $B(Q)$ should be significantly larger than the training loss $\hat{l}(Q,\, S)$. More specifically, we expect that $\sum Q_i\Gamma_i$ is comparable to $\sum Q_i \hat{l}(c_i,\, S)$ and that both of these quantities are significantly larger than zero. In order for $\sum Q_i\Gamma_i$ to be significantly larger than zero we must have that a significant fraction of the posterior is such that Γ_i is significantly larger than zero, i.e., that $\ln\frac{Q_i}{P_i}$ is comparable to m and hence much larger than α. A concept for which $\ln\frac{Q_i}{P_i} \gg \alpha$ will be called a *typical posterior concept.* We now assume that the posterior is dominated by such typical posterior concepts.

The requirements that $Q_i \geq 0$ and that Q is pruned, i.e., $Q_i > 0$ implies $Q_i \geq P_i$, do not restrict the value of Q_i for typical posterior concepts. Here we focus on computing the optimal posterior probability of typical posterior concepts using only the constraint that $\sum Q_i = 1$. We use the technique of Lagrange multipliers. The first step is to compute $\frac{\partial B}{\partial Q_i}$.

$$\begin{aligned}
B(Q) &= \sum Q_i(\hat{l}_i + \Gamma_i) \\
\frac{\partial B}{\partial Q_i} &= \hat{l}_i + \Gamma_i + Q_i\frac{\partial \Gamma_i}{\partial Q_i} \\
\frac{\partial \Gamma_i}{\partial Q_i} &= \frac{1}{2\Gamma_i\beta Q_i} \\
\frac{\partial B}{\partial Q_i} &= \hat{l}_i + \Gamma_i + \frac{1}{2\beta\Gamma_i}
\end{aligned}$$

We now note that if c_i is a typical posterior concept then $\Gamma_i \gg \frac{1}{2\beta\Gamma_i}$. More specifically we have the following.

$$\begin{aligned}
\frac{\Gamma_i}{\frac{1}{2\beta\Gamma_i}} &= 2\beta\Gamma_i^2 \\
&= 2(\ln\frac{Q_i}{P_i} + \alpha) \\
&\gg 1
\end{aligned}$$

So for any typical posterior concept c_i we have the following.

$$\frac{\partial B}{\partial Q_i} \approx \hat{l}_i + \Gamma_i \tag{9}$$

Formula 9 implies that, for typical posterior concepts, $\frac{\partial B}{\partial Q_i} > 0$. So we can reduce $B(Q)$ by moving mass from typical concept c_i to typical concept c_j provided $\frac{\partial B}{\partial Q_i} > \frac{\partial B}{\partial Q_j}$. Furthermore, formula 9 implies that for typical concepts we have that $\frac{\partial B}{\partial Q_i}$ gets smaller as Q_i gets smaller. Hence this process of reducing $B(Q)$ stabilizes when we have $\frac{\partial B}{\partial Q_i} = \frac{\partial B}{\partial Q_j}$ for all typical posterior concepts c_i and c_j. So for the optimal posterior there must exist a value z such that for all typical posterior concepts c_i we have the following.

$$\begin{aligned} \frac{\partial B}{\partial Q_i} &\approx \hat{l}_i + \Gamma_i \approx z \\ \Gamma_i &\approx z - \hat{l}_i \\ \ln \frac{Q_i}{P_i} + \alpha &\approx \beta(x - \hat{l}_i)^2 \\ Q_i &\approx P_i e^{2m(z-\hat{l}_i)^2} \end{aligned} \qquad (10)$$

Assuming that the majority of mass of the optimal posterior is at typical posterior concepts it is not very important what probability we assign to atypical concepts provided that we use a distribution approximately satisfying formula 10 for typical concepts. Therefore it seems that a good approximation of the optimal posterior is given by formula 2 which we repeat here for convenience.

$$Q_i^* \approx \begin{cases} 0 & \text{if } \hat{l}_i > z \\ P_i\alpha & \text{if } \hat{l}_i = z \\ P_i e^{2m(z-\hat{l}_i)^2} & \text{if } \hat{l}_i < z \end{cases}$$

The parameters z and α can be defined as the unique values in [0, 1] such that $\sum Q_i = 1$. More concretely, these parameters can be defined as follows.

$$\begin{aligned} w(x) &= \sum_{\hat{l}_i < x} P_i e^{2m(x-\hat{l}_i)^2} \\ z &= \sup\{x : w(x) \leq 1\} \\ \alpha &= \frac{1 - w(z)}{\sum_{\hat{l}_i = z} P_i} \end{aligned}$$

5 The Finite-Variance Case

A common objection to the PAC-Bayesian framework is that it applies only to bounded loss functions while in classical Bayesian inference one is interested in (typically unbounded) log loss. Here we try to address this objection by giving a PAC-Bayesian model selection algorithm that can be used with unbounded log loss. The "result" of this section is not mathematically deep — it is achieved by simply replacing a finite-variance quantity with a bounded quantity with bounds derived from the mean and variance of the original. This simple trick, although mathematically trivial, seems to provide insight into the relationship between the PAC-Bayesian and classical Bayesian approach.

A loss function L will be said to be *finite-variance* with respect to the underlying distribution if for each concept c_i the quantity $L(c_i, x)$ has finite mean and variance (over the choice of x). For any finite-variance loss function L we define the following bounded version of L where $\overline{L_i}$ and σ_i are the mean and variance, over the choice of x, of $L(c_i, x)$ and where w is a "width" parameter which is the same for all concepts.

$$\mathcal{L}_w(c_i, x) = \max(\overline{L_i} - w\sigma_i, \min(\overline{L_i} + w\sigma_i, L(c_i, x)))$$

By Chebychev's inequality we have that for any given c_i, with probability at least $1 - (\frac{1}{w})^2$ over the choice of x, $\mathcal{L}_w(c_i, x) = L(c_i, x)$. We now have that $\frac{1}{2} + \frac{\mathcal{L}_w(c, x) - \overline{L_i}}{2w\sigma_i}$ is in the interval [0, 1]. By inserting the loss function $\frac{1}{2} + \frac{\mathcal{L}_w(c_i, x) - \overline{L_i}}{2w\sigma_i}$ into (1) one can prove the following.

$$\forall^\delta S \;\forall c_i \;\; \mathcal{L}_w(c_i) \leq \hat{\mathcal{L}}_w(c_i, S) + \frac{\sigma_i}{\sqrt{m}} w \sqrt{2(\ln \frac{1}{P_i} + \ln \frac{1}{\delta})} \qquad (11)$$

Note that the quantity $\frac{\sigma_i}{\sqrt{m}}$ is the standard deviation of the measurement $\hat{L}(c, S)$. PAC-Bayesian model selection based on the bound in (11) involves a dependence on σ_i not found in either classical Bayesian model selection or in earlier PAC-Bayesian model-selection formulas.

Kearns et al. [6] observe that standard PAC risk minimization formulas put more weight on model complexity than does classical Bayesian inference or MDL. Note that if σ_i is sufficiently small, i.e., if $\hat{\mathcal{L}}(c, S)$ is insensitive to the choice of S, then (11) puts more weight on $\hat{\mathcal{L}}(c, S)$ than on $\frac{\ln \frac{1}{P_i}}{m}$. The possibility of $\hat{\mathcal{L}}(c, S)$ receiving more weight than $\frac{\ln \frac{1}{P_i}}{m}$ distinguishes (11) from earlier risk minimization formulas.

Formula (11) can be viewed as a finite-variance version of formula (1). It should also possible to derive finite-variance analogous of theorem 1 and lemma 2. Let l be the bounded loss function $\frac{1}{2} + \frac{\mathcal{L}_w(c_i, x) - \overline{L_i}}{2w\sigma_i}$. Formula (5) states that, with probability at least $1 - \delta$, when a sample S is taken we have the following.

$$\sum P_i e^{(2m-1)\gamma_i^2} \leq \frac{4m}{\delta} \qquad (5)$$

In (5) we now have that γ_i equals $\frac{|\mathcal{L}_w(c_i) - \hat{\mathcal{L}}_w(c_i, S)|}{2w\sigma_i}$. To prove a finite-variance version of lemma 2 we need to bound the quantity $\mathcal{L}_w(Q) - \hat{\mathcal{L}}_w(Q, S)$. This quantity is now bounded by $2w \sum Q_i\sigma_i\gamma_i$. We can now maximize the quantity $\sum Q_i\sigma_i\gamma_i$ over the possible values of γ_i satisfying (5). By an argument similar to that of section 3 it should then be possible to show that for $\ln \frac{1}{\delta} \leq 2m$ we have that with probability at least $1 - \delta$ we have the following for all pruned distributions Q with $D(Q||P) \leq 2m$.

$$\begin{aligned} L_w(Q) &\leq \hat{L}_w(Q, S) + B(Q) \\ B(Q) &\approx \sum Q_i \frac{\sigma_i}{\sqrt{m}} w \sqrt{2 \ln \frac{Q_i}{P_i}} \end{aligned}$$

Using methods similar to those of section 4 it should then be possible to show the following.

$$Q_i^* \approx \begin{cases} 0 & \text{if } \hat{L}_i > z \\ P_i \alpha & \text{if } \hat{L}_i = z \\ P_i e^{\frac{m(z-\hat{L}_i)^2}{2w^2\sigma_i^2}} & \text{if } \hat{L}_i < z \end{cases} \quad (12)$$

As in formula 2, the parameters z and α are selected so that $\sum Q_i = 1$.

6 Conclusion

Model averaging has important applications in a variety of machine learning applications including language modeling and decision trees. This paper provides a PAC upper bound on the generalization error of weighted model mixtures. It remains to be seen whether the particular posterior distributions of formulas 2 and 12 prove to be empirically useful. Whether or not these weighting formulas prove useful, the results of this paper show that model averaging can be given a theoretical PAC justification independent of any Bayesian assumptions.

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